

StudyAbroad.AI — Master Project Plan & Strategy

Autonomous Multi-Agent Global Education Platform (12 Autonomous AI Agents)

System: StudyAbroad.AI
Platform

Architecture: 12 Autonomous
Agents

Distribution: Confidential Client
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🌐 StudyAbroad.AI — Project Plan Document

Fully Autonomous Study Abroad Intelligence Platform

Version: 1.0 | **Date:** September 2026 | **Status:** Planning

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1. Vision & Problem Statement

The Big Idea

A **single autonomous platform** that replaces: study abroad agencies (-), manual research (weeks of work), document consultants (/SOP), and scholarship hunting (months of confusion).

From 'I think I want to study abroad' to 'I have graduated, found a job, and got my PR' — every single step, fully guided and automated by AI.

Problems We Solve

Problem Today	Our AI Solution
Students pay to agents	Free AI that does 10x better
Outdated PDF guides	Real-time scraped, always fresh data
Takes 6 months of research	10-minute AI profile gives instant plan
Miss deadlines and scholarships	Autonomous agents monitor 24/7
Bad SOP gets rejected	AI writes and scores your SOP in your own voice
No idea which professor to contact	AI finds and drafts the email
Visa rejection due to wrong docs	AI guides step by step
Scholarship scams	AI validates every scholarship
No post-arrival support	Platform follows you until PR
No career outcome data	AI shows ROI of every university choice

What We Replace

Service Replaced	Their Cost	Our Advantage
Study Abroad Agency	-,000	Free + 24/7 + no bias
SOP Consultant	-	Instant + personalized per university
Scholarship Advisor	+	Real-time scraped + niche matching
Visa Consultant	-	Country-specific + always updated
Career Counselor	/session	Data-driven ROI + salary projections

2. Market Analysis

Market Size

- 6+ million students study abroad annually
- industry dominated by expensive human consultants
- 1.5B non-native English speakers need guidance
- Study abroad numbers growing 8% year-over-year

Competitor Gaps

Competitor	What They Do	What They Miss
Leverage Edu	Human-led + AI tools	Expensive, biased toward partner universities
Yocket	Community + university search	No document generation, no agents
AbroBot	AI university matching	Very limited, no full pipeline
All competitors	Pre-application only	Nobody covers post-arrival life

3. The 12 Autonomous AI Agents

Agent 1 — UniversityScraperAgent

Autonomously crawls and maintains live university data globally

- 600+ universities, 8,000+ specific programs
- Countries: USA, UK, Canada, Germany, Australia, Netherlands, Sweden, France, Switzerland, New Zealand, Ireland, Japan, South Korea, Singapore, UAE, Finland, Denmark, Italy, Spain
- Data: programs, tuition, admission requirements, deadlines, acceptance rates, scholarships, intake schedules
- Re-scrapes weekly with change detection

Agent 2 — ScholarshipHunterAgent

Monitors 100+ scholarship sources globally, 24/7

- Chevening, Fulbright, DAAD, Erasmus+, Commonwealth, MEXT, KAIST, CSC, Australia Awards, Vanier, Swedish Institute
- Elite: Gates Cambridge, Rhodes, Marshall, Knight-Hennessy
- University-specific: TA/RA positions, tuition waivers, fellowships
- Hidden gem discovery: less competitive, highly relevant scholarships

Agent 3 — RecommenderAgent (Core Brain)

Builds personalized university strategy using RAG + semantic search

- Dream / Target / Safety school categorization
- Match score (0–100%) with detailed reasoning
- Admission probability estimate
- Total cost-of-attendance breakdown
- 5–university comparison matrix

Agent 4 — DocumentCraftingAgent (Voice-Preserved)

Writes ALL application documents in the student's own voice

- SOP (per university + program + professor)
- Personal Statement, Research Proposal, LOR drafts
- Academic CV, Diversity Statement, Scholarship Essays
- Voice Preservation: AI learns student's writing style via 10–question session
- Quality checks: plagiarism, AI-detection score, clarity, fit score

Agent 5 — ProfessorOutreachAgent

Finds funded PhD supervisors and drafts hyper-personalized outreach

- Scrapes professor research interests, publications, active grants
- Semantic matching: student research background vs professor work
- Generates personalized cold emails and follow-up sequences
- Tracks: sent / opened / replied / interview scheduled

Agent 6 — DeadlineWatcherAgent

Ensures zero missed deadlines across the entire journey

- Tracks: application, scholarship, visa, housing, post-admission immigration deadlines
- Alerts via: in-app, email, WhatsApp, Telegram, Google Calendar sync
- Weekly auto-generated task list
- Smart priority: urgent + high-impact tasks first

Agent 7 — VisaGuideAgent

Complete visa journey from country decision to passport stamp

- Country-specific roadmaps: F-1 USA, Student Visa UK, German Study Visa, Canadian Study Permit, Australian Visa 500
- Bank statement calculator (exact formula per country)
- Visa interview mock: AI plays visa officer
- Visa rejection recovery: analysis + appeal plan
- Post-study PR pathway guide per country

Agent 8 — StudyAbroadCounselorAgent

24/7 AI counselor with full memory of student journey

- Full context-aware answers using complete student profile
- Remembers every conversation and preference
- Escalation to verified human experts for edge cases
- Multilingual: responds in student's native language

Agent 9 — CareerROIAgent (Key Differentiator)

Data-driven career and financial outcome analysis

- Median salary 1yr/3yr/5yr post-graduation per program
- Top employers hiring from each university
- ROI calculator: tuition cost / salary boost = years to break even
- Immigration pathways: which degrees in which country open best PR routes
- Best country for your field comparison tool

Agent 10 — ScamDetectorAgent

Protects students from scholarship scams and unaccredited institutions

- Validates every scholarship against official databases
- Flags: unofficial domains, payment requests, unrealistic awards
- University accreditation verification
- Community-reported scam database

Agent 11 — FinancialDocAgent

Full financial documentation support for visa and application requirements

- Exact bank balance calculation per country per visa type
- Bank Statement Preparation Guide for home country banking systems
- Handles banking restriction contexts (Bangladesh, Pakistan, Nigeria, etc.)
- Education loan comparison tool

Agent 12 — WellbeingCompanionAgent

Mental health support and social integration

- Culture shock preparation guide per country
- First 30/60/90 day plan post-arrival
- Anonymous mental health check-in system
- Connect with students from same home country at target university

4. Complete Feature Map (50+ Features)

Discovery and Research

- Natural language university search
- Program-level granular data
- Subject-specific rankings (CSRankings, Shanghai Subject)
- Side-by-side comparison (up to 5 universities)
- Cost of living calculator per city (Numbeo)
- Student review aggregator (Reddit, TheStudentRoom, GMAT Club)
- Alumni employment outcomes per program
- Safety index per country/city

Profile and Eligibility

- Comprehensive profile wizard (academic, financial, personal, research)
- GPA converter: 4.0 scale, percentage, CGPA, UK classification
- Transcript upload: AI auto-extracts GPA and grades
- LinkedIn import: auto-fills work and research experience
- Test score tracker: IELTS, TOEFL, GRE, GMAT, LSAT, MCAT
- AI eligibility checker with instant reasoning
- Profile strength score (0–100) + improvement roadmap
- What-if simulator: score improvement vs. new university options
- Peer benchmark against admitted students at target universities

Document Preparation

- Voice preservation system (10-question style profiler)
- SOP generator (university + program + professor specific)
- SOP quality scorer (Clarity, Motivation, Fit — rated 0–10 each)

- Personal statement, Research proposal, LOR drafts
- Academic CV builder, Scholarship essay writer
- Plagiarism + AI-detection score checker
- Document vault with version history and collaboration
- One-click re-tailoring from one university to another

Application Management

- Kanban board tracker (Research to Decision)
- University-specific checklists
- Application fee tracker and budget calculator
- Offer letter AI analyzer
- Multi-offer financial comparison
- Admission interview prep with AI mock

Scholarships and Funding

- Personalized scholarship matching engine
- Hidden gem scholarship discovery
- Funding gap calculator
- TA/RA opportunity finder
- Student loan comparison by country
- Scholarship essay bank (anonymized winning examples)
- Scholarship scam detector

Visa and Immigration

- Visa roadmap generator per country
- Bank statement requirement calculator
- Visa interview mock coach
- Visa rejection recovery plan
- Post-study PR pathway guide
- Immigration policy change alerts

Pre-Departure and Arrival

- Pre-departure checklist per country
- Health insurance guide
- Banking setup guide per country
- Verified housing finder (scam-free)
- First 30/60/90 day plan generator
- Cultural adaptation guide

Life Abroad (Post-Arrival)

- Student community connector

- Part-time job board (within visa work hour limits)
- Mental health check-in system
- Post-graduation job board
- OPT/PGWP/PSW application timeline tracker
- Alumni network access

Analytics and Intelligence

- Application success predictor
 - Admission trend charts (5 years)
 - Salary benchmarks post-graduation
 - ROI comparison across shortlisted universities
 - Immigration path probability scores
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5. Technical Architecture

FRONTEND LAYER

- Web App: Next.js 14 (App Router, SSR, SEO)
- Mobile: React Native + Expo (iOS + Android)
- UI: Tailwind CSS + shadcn/ui + Framer Motion
- Real-time: Socket.io
- Multilingual: i18next (10 languages)

API GATEWAY LAYER (FastAPI)

- Auth Service: NextAuth.js + Google + LinkedIn OAuth + JWT
- Profile Service: CRUD, import, scoring
- Search and Recommendation Service: RAG endpoint
- Document Service: generation, scoring, vault
- Agent Orchestration: LangGraph + CrewAI hybrid
- Notification Service: multi-channel dispatch
- Analytics Service: dashboards, reports

AI AGENT LAYER

- UniversityScraperAgent [CrewAI pipeline]
- ScholarshipHunterAgent [CrewAI pipeline]
- RecommenderAgent [LangGraph state machine]
- DocumentCraftingAgent [LangGraph + voice model]
- ProfessorOutreachAgent [CrewAI pipeline]
- DeadlineWatcherAgent [Cron + LangGraph]
- VisaGuideAgent [LangGraph state machine]
- CounselorAgent [LangGraph + memory]
- CareerROIAgent [CrewAI + data scraping]
- ScamDetectorAgent [Rule-based + LLM validation]
- FinancialDocAgent [LangGraph + templates]
- WellbeingCompanionAgent [LangGraph + sentiment analysis]

TRUST LAYER

- Human-in-the-Loop (HITL): expert review for flagged advice
- Verified Expert Network: human counselors on standby
- Confidence Scoring: AI shows uncertainty + sources
- Audit Logs: every agent action logged for transparency

DATA LAYER

- PostgreSQL + Prisma: users, profiles, applications
- ChromaDB to Pinecone: university and scholarship vectors
- Redis + Celery: job queues, caching
- S3 / Cloudinary: encrypted document storage
- APScheduler: weekly scraping cron jobs
- TimescaleDB: time-series for trend analytics

OBSERVABILITY AND SECURITY LAYER

- Langfuse: AI agent tracing, token usage, cost monitoring
 - LangSmith: LLM evaluation and regression testing
 - Sentry: error tracking and performance monitoring
 - AES-256 encryption: all uploaded documents
 - GDPR compliance: data deletion, export, consent flows
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6. Tech Stack

Layer	Technology	Reason
Web Frontend	Next.js 14 (App Router)	SEO + SSR + Full-stack React
Mobile	React Native + Expo	iOS + Android from one codebase
UI Components	Tailwind CSS + shadcn/ui	Fast, beautiful, accessible
Animations	Framer Motion	Premium micro-interactions
Backend API	FastAPI (Python)	Async, fast, ideal for AI workloads
Agent Framework	LangGraph + CrewAI (hybrid)	LangGraph for stateful; CrewAI for pipelines
LLM	Gemini 1.5 Pro + Flash / Groq	Free tier, powerful, multimodal
Embeddings	Google text-embedding-004	Best free embeddings available
Vector DB	ChromaDB (dev) to Pinecone (prod)	Scalable semantic search
Web Scraping	Crawl4AI + Playwright + BeautifulSoup	AI-native, handles dynamic JS pages
Relational DB	PostgreSQL + Prisma ORM	Production-grade, type-safe
Cache/Queue	Redis + Celery	Background agent jobs, rate limiting
Auth	NextAuth.js + Google + LinkedIn OAuth	Easy login + profile import
Document Storage	Cloudinary + AES-256	Secure, scalable file vault
Email Alerts	Resend (3000/mo free)	Transactional emails
WhatsApp Alerts	Twilio WhatsApp API	Deadline alerts on mobile
Real-time	Socket.io	Live notifications and chat
Multilingual	i18next	10+ language platform support
Deployment	Vercel (web) + Railway (API + agents)	Free tiers, fast deploy
AI Monitoring	Langfuse + LangSmith	Full agent tracing and evaluation
Error Tracking	Sentry	Production stability
User Analytics	PostHog (free)	User behavior and funnels

7. 8-Phase Build Roadmap

Phase 1 — Foundation (Week 1)

- Monorepo setup (Next.js + FastAPI)
- PostgreSQL schema design

- Auth system: Google + LinkedIn OAuth + JWT
- Student profile wizard UI
- Basic navigation shell
- CI/CD pipeline

Phase 2 — Data Engine (Week 2)

- UniversityScraperAgent build
- ScholarshipHunterAgent build
- ChromaDB vector store setup
- Embedding pipeline: chunk to embed to store
- Seed 200 universities + 500 scholarships across 15 countries
- Admin dashboard for scraping jobs

Phase 3 — Core AI (Week 3)

- RAG pipeline (semantic search)
- RecommenderAgent
- CareerROIAgent
- University Explorer UI
- Scholarship Hub UI

Phase 4 — Document Suite (Week 4)

- Voice Preservation System (10-question profiler)
- DocumentCraftingAgent (SOP, LOR, CV, Essays)
- Document editor with inline AI suggestions
- SOP quality scorer
- Plagiarism + AI-detection checker
- Document Vault

Phase 5 — Agent Automation (Week 5)

- ProfessorOutreachAgent
- DeadlineWatcherAgent (email, WhatsApp, calendar)
- VisaGuideAgent
- FinancialDocAgent
- ScamDetectorAgent
- Application Tracker (Kanban)

Phase 6 — Intelligence Layer (Week 6)

- CounselorAgent (24/7 chat with memory)
- WellbeingCompanionAgent
- Human-in-the-Loop trust layer
- What-if simulator

- Analytics dashboard

Phase 7 — Mobile and Multilingual (Week 7)

- React Native mobile app
- Push notifications
- i18next (10 languages)
- Real-time notifications via Socket.io
- Post-arrival Life Abroad module

Phase 8 — Security, Polish and Launch (Week 8)

- AES-256 document encryption
- GDPR compliance
- Langfuse + Sentry monitoring
- SEO optimization
- Landing page and onboarding flow
- Production deployment
- Beta tester recruitment (100 students)

8. Competitor Comparison

Feature	Leverage Edu	Yocket	AbroBot	StudyAbroad.AI
12 Autonomous AI Agents	No	No	No	YES
Voice-Preserved Documents	No	No	Partial	YES
Career ROI and Salary Data	No	Partial	No	YES
Scholarship Scam Detection	No	No	No	YES
Visa Rejection Recovery	No	No	No	YES
Post-Arrival Life Support	No	No	No	YES
Professor Outreach Agent	No	No	No	YES
Financial Document Guidance	Partial	No	No	YES
Mobile App	Yes	Yes	No	YES
Free Core Features	No	Partial	Partial	YES
Multilingual 10 languages	No	No	No	YES
Human-in-the-Loop Trust Layer	Yes (expensive)	No	No	YES (free)

9. Portfolio Value

AI/ML Skill	Level Shown
Multi-Agent AI (LangGraph + CrewAI)	Expert
RAG Pipeline with Reranking	Expert
Autonomous Web Scraping	Advanced
Vector Database Design	Expert
LLM Prompt Engineering	Expert
Full-Stack (Next.js + FastAPI)	Advanced
Mobile Development (React Native)	Intermediate
Real-time Systems (WebSocket)	Advanced
Production AI Monitoring	Expert
System Design at Scale	Expert
Security and GDPR Architecture	Intermediate
MLOps and Production Deployment	Advanced

10. Business Model

Tier	Price	Included
Free	/month	University search, basic recommendations, 1 SOP draft/month
Pro	/month	Unlimited documents, all 12 agents, professor outreach
Premium	/month	Expert human review, priority support, full ROI analytics
Agency	/month	White-label, bulk student management

Additional Revenue

- University Partnerships: -,000/month for featured placement
- Affiliate Revenue: IELTS prep, student housing, insurance, banking
- Data API: sell verified university/scholarship data to EdTech platforms
- Expert Marketplace: 15% commission on human counselor sessions

Year 1 Target

10,000 users x 20% Pro conversion x /month = ,000 ARR

